Review about optical computing

Abstract

Optical computing aims to use light to perform core operations such as multiplication, addition, and matrix multiplication. This review focuses on the mainstream AI-oriented direction of optical computing: constructing optical systems that mimic electronic circuits or deep-learning-based neural-network structures. Other active directions, such as photonic Ising machines, reservoir computing, inverse design based computing component and photonic quantum computing, are acknowledged but not emphasized here; they will be explored in depth when needed.

For AI-related applications, three main approaches are considered: photonic neural networks, diffractive neural networks, and photonic AI accelerators. Photonic neural networks implement deep-learning models on chip and mainly target matrix multiplication, which is typically the most power-hungry operation in modern AI workloads. According to operation principle, two major routes are common. The first is coherent-light computing based on MZI meshes and typically single-wavelength operation. A single MZI unit can realize an SU(2) transformation, and properly arranged meshes can implement an $N \times N$ unitary matrix transform. This method avoids large multi-laser sources, but it is strongly affected by phase noise and therefore requires frequent, delicate tuning of relative phase. The second route is incoherent-light computing based on micro-ring-resonator (MRR) arrays and wavelength-division multiplexing (WDM). Different wavelengths represent orthogonal channels, and weights are tuned by shifting the resonant peaks of MRRs to control transmission. This approach is often more stable and easier to construct than coherent meshes, but it relies on many wavelengths and needs closed-loop control because MRRs are sensitive to fabrication variation and environmental drift. Noticebaly, some reseaches also use MZI mesh for incohrent computing, but the overall logic of implementation is quite similar to MRR-based computing, namely adding the intensity of different wavelength at the final stage to realize addition.

In contrast, diffractive neural networks normally deploy computation in free space. From a Fourier-optics perspective, free-space propagation behaves as a convolution-like mixing process, although the corresponding kernel cannot be tuned arbitrarily. A typical diffractive network uses cascaded phase screens separated by designed propagation distances; performance is tuned via phase-modulation depth and spacing. This approach is relatively easy to prototype (for example, using several spatial light modulators), but it often has lower accuracy and is harder to fine tune than on-chip photonic neural networks. Since diffraction is a universial phenomean for wave, so this idea can also be implemented on chip using slab region.

For both photonic neural networks and diffractive neural networks, linear operations dominate. Because light–matter interaction is weak, fully optical nonlinear activation is often not cost-effective and cannot be tuned as flexibly as electronic activations. Therefore, many published systems introduce electronic nonlinear stages or rely on square-law photodetection. A third path, photonic AI acceleration, treats photonics as a high-speed linear-computing block instead of building a fully photonic end-to-end neural network. These systems focus on accelerating the most time-consuming parts of neural workloads, especially fully connected layers and matrix multiplication. The following sections review representative papers under these three ideas and focus on three questions for each work: what is genuinely new, how the device or algorithm works, and how the design is actually used to perform computation.

1 Photonic neural network

1.1 MZI-based coherent PNN

The papers in this subsection share the same basic workflow: encode the input vector in optical field amplitudes and phases, use a programmable MZI mesh to realize a linear transformation, and read the result after optical propagation. To avoid repetition, I do not re-explain the same MZI basics in every paper. Instead, each entry focuses on what that paper adds, how its specific design works, and how the hardware is actually used.

1.1.1 Deep learning with coherent nanophotonic circuits

Authors: Yichen Shen et al.
Journal: Nature Photonics 11(7), 441–446 (2017)
DOI: 10.1038/nphoton.2017.93

Innovation. This paper is the starting point of modern coherent integrated optical neural networks. Its key contribution is to show that a programmable MZI mesh can implement the linear layer of a neural network in hardware, not just a simple switch or interferometer experiment.

Working principle. The central mathematical tool is singular value decomposition, $M = U\Sigma V^\dagger$. A general matrix is difficult to build directly in optics, so the paper breaks it into two unitary matrices and one diagonal matrix. In a low-loss coherent circuit, one MZI acts as a tunable 2×2 unitary mixer: the two couplers split and recombine light, while the internal phase shifters set how much optical amplitude goes to each output and what phase it carries. When many MZIs are arranged into a mesh, the first mesh can implement $V^\dagger$, a diagonal stage can implement Σ by setting channel-by-channel attenuation, and a second mesh can implement U. The input data are encoded in the complex optical amplitudes launched into the input waveguides, and the trainable weights are encoded in the phase settings of the meshes and in the diagonal attenuation values. Once these settings are loaded, the chip performs the matrix multiplication physically as the optical field propagates through the circuit.

How to use it. In practice, one first trains or chooses the target weight matrix in software, decomposes it into $U\Sigma V^\dagger$, converts those three pieces into MZI phase settings and diagonal attenuation values, programs the chip, and then injects the input vector as optical amplitudes. The optical outputs are the multiplied result and can be sent to the next layer or to detectors. The paper demonstrated this workflow with a 56-MZI silicon photonic processor on a vowel-recognition task, which is why it became the reference design for later coherent ONN work.

1.1.2 Training of photonic neural networks through in situ backpropagation and gradient measurement

Authors: Tyler W. Hughes, Momchil Minkov, Yu Shi, and Shanhui Fan
Journal: Optica 5(7), 864–871 (2018)
DOI: 10.1364/OPTICA.5.000864

Innovation, working principle, and use. This paper answered a crucial question left open by early ONN demonstrations: how can the chip measure the gradient of its own phase shifters instead of being trained entirely offline in a digital computer. The circuit still performs a linear optical transform with programmable MZI meshes, but the innovation is the learning rule. Using the adjoint-variable method, the authors showed that the derivative of the loss with respect to one optical parameter is proportional to the overlap between two fields inside the chip: the forward field generated by the input signal and an adjoint field generated by sending the output error backward

through the same reciprocal circuit. Physically, this means the processor can estimate a gradient by launching a normal inference signal, launching an error signal from the output side, and reading intensity changes at monitors or interference points. The important point for beginners is that the chip is not "guessing" how to update itself; it is exploiting reciprocity so the same optical hardware that computes the forward pass also contains the information needed for backpropagation. This paper therefore turned in situ training from a vague goal into a concrete measurement framework for photonic neural networks.

1.1.3 Experimentally realized in situ backpropagation for deep learning in photonic neural networks

Authors: Shiyang Pai, Tyler W. Hughes, Momchil Minkov, Ian A. D. Williamson, Momchil Milanizadeh, Niels C. Harris, David A. B. Miller, and Shanhui Fan
Journal: Science 380(6643), 398–404 (2023)
DOI: 10.1126/science.ade8450

Innovation, working principle, and use. This paper is the experimental realization of the earlier adjoint-gradient theory. The hardware is a programmable silicon photonic network in which MZI meshes implement the linear layers, while monitors and reciprocal optical paths are added so the chip can access gradient information during training. The procedure is to send the normal input forward, generate an error signal from the output side, and let the forward and backward fields interfere inside the same physical layer. From the measured local intensities, the gradient of each phase shifter can be extracted directly on hardware, and an electronic controller then updates the phase settings. For a reader without an optics background, the simplest interpretation is that the chip reuses its own wave propagation to compute both the inference result and the learning signal, instead of repeatedly perturbing one weight at a time or relying on a separate digital twin. That is the specific breakthrough of this work: it converted in situ backpropagation from a theoretical proposal into a working silicon-photonic training experiment.

1.1.4 Matrix Optimization on Universal Unitary Photonic Devices

Authors: Sunil Pai, Ben Bartlett, Olav Solgaard, and David A. B. Miller
Journal: Physical Review Applied 11, 064044 (2019)
DOI: 10.1103/PhysRevApplied.11.064044

Innovation, working principle, and use. This paper did not mainly propose a new photonic computing primitive, but instead addressed a fundamental training problem in universal MZI-based unitary meshes. The key observation is that standard uniform-random initialization is a poor match to large rectangular interferometer meshes: because the device is only locally connected, light propagation behaves like a random walk, so the implemented matrix becomes increasingly *banded* as the mesh grows, with weak long-range input-output coupling. That makes gradient-based optimization of arbitrary random unitary targets slow or unsuccessful. To solve this, the paper derived the required physical parameter distribution for a Haar-random unitary directly in terms of the mesh MZI transmissivities and phases, showing that the more central MZIs must be biased close to the cross state and also become much more sensitive to error. Based on this, the authors proposed *Haar initialization*, which initializes each MZI from the correct random-unitary distribution instead of from naive random phases, so that light from any input is spread more uniformly across outputs from the start. They then proposed two architectural improvements to further ease optimization: a *redundant rectangular mesh* with extra tunable MZI layers, and a *permuting rectangular mesh* that inserts permutation layers so distant waveguides can interact

more effectively. Using TensorFlow simulations for meshes up to $N = 128$, they showed that Haar initialization strongly improves convergence, that permuting meshes reduce off-diagonal error and relax the local-connectivity bottleneck without increasing parameter count, and that redundant meshes can reduce convergence error by up to five orders of magnitude and shorten optimization time by around two orders of magnitude.

1.1.5 Efficient Training and Design of Photonic Neural Network through Neuroevolution

Authors: Tian Zhang, Jia Wang, Yihang Dan, Yuxiang Lanqiu, Jian Dai, Xu Han, Xiaojuan Sun, and Kun Xu
Journal: Optics Express 27(26), 37150–37163 (2019)
DOI: 10.1364/OE.27.037150

Innovation, working principle, and use. This paper focused on the *training strategy* of photonic neural networks rather than on proposing a new photonic device structure. The main idea was to use *neuroevolution* as an alternative to conventional gradient-based learning for integrated ONNs, where the trainable weights are implemented by optical phase shifters. Instead of relying on backpropagation through an exact hardware model, the authors treated ONN design and training as a population-based optimization problem and introduced two typical neuroevolution algorithms—genetic algorithm (GA) and particle swarm optimization (PSO)—to optimize both network hyperparameters and connection weights. In this way, the method is attractive for photonic systems whose internal device responses may be difficult to model accurately or differentiate analytically. To verify the idea, the paper applied the trained ONNs to several classification tasks, including the iris plants dataset, the wine recognition dataset, and optical modulation-format recognition. The reported results showed that the neuroevolution-based training achieved competitive accuracy and stability compared with more traditional learning approaches, suggesting that black-box evolutionary optimization can be a feasible route for ONN training when model-based gradients are inconvenient or unreliable.

1.1.6 A Single Layer Neural Network Implemented by a 4×4 MZI-Based Optical Processor

Authors: Farhad Shokraneh, Simon Geoffroy-Gagnon, Mohammadreza Sanadgol Nezami, and Odile Liboiron-Ladouceur
Journal: IEEE Photonics Journal 11(6), 1–12 (2019)
DOI: 10.1109/JPHOT.2019.2952562

Innovation, working principle, and use. This paper demonstrated a *single-layer optical neural network* implemented on a reconfigurable 4×4 MZI-based optical processor. Instead of proposing a large deep ONN, it focused on how to program a compact interferometric mesh so that a trained neural-network weight matrix could be physically realized on chip. The processor consists of an SU(4) interferometric section built from six MZIs, followed by a diagonal matrix multiplication section for controlling the output amplitudes, so the overall device can implement a linear transformation matrix and can also be viewed as a building block for larger SVD-based processors. To make the hardware usable, the authors first calibrated the individual MZIs to compensate fabrication-induced phase offsets, then trained a single-layer classifier offline on a synthetic linearly separable four-dimensional Gaussian dataset, and finally converted the trained weight matrix into the required phase settings for the chip. An important point is that they intentionally did *not* use a nonlinear activation function in this work, because for a single-layer classifier with a monotonic nonlinearity

the class decision is unchanged; the emphasis here was therefore on the practical realization of the linear photonic layer itself. In this sense, the contribution of the paper is less about network depth and more about showing a full pipeline from matrix design, calibration, and phase extraction to experimental programming of a fabricated MZI processor for neural-network-style classification.

1.1.7 Reprogrammable Electro-Optic Nonlinear Activation Functions for Optical Neural Networks

Authors: Ian A. D. Williamson, Tyler W. Hughes, Momchil Minkov, Ben Bartlett, Sunil Pai, and Shanhui Fan
Journal: IEEE Journal of Selected Topics in Quantum Electronics 26(1), 7700412 (2020)
DOI: 10.1109/JSTQE.2019.2930455

Innovation, working principle, and use. This paper targeted the part that coherent MZI meshes cannot provide by themselves: a strong and programmable nonlinear activation. The linear optical layer still performs matrix multiplication, but a neural network also needs a nonlinear step after each layer. Instead of relying on weak all-optical material nonlinearities, the authors proposed an electro-optic optical-to-optical activation. A small tap sends part of the optical signal to a photodetector, the photocurrent is amplified electrically, and that voltage drives a phase modulator placed in the main optical path, usually inside another MZI. As a result, the input optical power changes the transmission of its own output path, so the block behaves like an activation function while leaving most of the signal in the optical domain. Because the electrical gain and bias can be changed, the same device can approximate different response curves such as saturating or ReLU-like functions. The real innovation is therefore not just "adding nonlinearity", but adding a tunable and practically strong nonlinearity that can be inserted between optical matrix multipliers on a chip.

1.1.8 Experimental Realization of Arbitrary Activation Functions for Optical Neural Networks

Authors: Monireh Moayedi Pour Fard, Ian A. D. Williamson, Matthew Edwards, Ke Liu, Sunil Pai, Ben Bartlett, Momchil Minkov, Tyler W. Hughes, Shanhui Fan, and Thien An Nguyen
Journal: Optics Express 28(8), 12138–12148 (2020)
DOI: 10.1364/OE.391473

Innovation, working principle, and use. This paper pushed the electro-optic activation idea from a mainly architectural concept toward *experimental realization*. Instead of relying on weak intrinsic optical nonlinearities, the authors demonstrated an *on-chip electro-optic circuit* that can implement arbitrary optical-to-optical nonlinear activation functions for optical neural networks. The core idea is to tap a small portion of the input optical signal, convert that portion into an electrical signal, process it electronically, and then use it to modulate the intensity of the remaining optical signal. Because the nonlinearity is produced through an electrical control path rather than through fixed material physics, the realized transfer curve is not restricted to conventional Kerr-like or saturable-optics line shapes. In this way, the same general activation hardware can synthesize a broad class of nonlinear responses, giving much more flexibility than purely optical nonlinear elements. The paper further argued, through system-level simulations, that such activation functions can improve ONN performance on tasks such as MNIST image classification. Another important point is that the architecture can realize nonlinear behavior with much lower optical signal attenuation than many traditional optical nonlinear approaches, which is especially valuable for building deeper cascaded ONNs where signal loss would otherwise accumulate rapidly. So the main contribution of the paper is not only to propose reconfigurable nonlinear activation, but to show that this

idea can be implemented in a practical on-chip electro-optic circuit rather than remaining a purely theoretical building block.

1.1.9 The Diamond Mesh, a Phase-Error- and Loss-Tolerant Field-Programmable MZI-Based Optical Processor for Optical Neural Networks

Authors: Farhad Shokraneh, Simon Geoffroy-Gagnon, and Odile Liboiron-Ladouceur
Journal: Optics Express 28(16), 23495–23508 (2020)
DOI: 10.1364/OE.395441

Innovation, working principle, and use. This paper proposed a *diamond-mesh* MZI topology for optical neural networks, targeting a practical weakness of conventional triangular Reck meshes. Compared with the Reck structure, the diamond mesh uses $(N-1)(N-2)/2$ additional reconfigurable MZIs, which makes the topology more symmetric and gives extra degrees of freedom during weight-matrix optimization in backpropagation. A key idea is that these additional MZIs also allow the processor to remove excess light through tapered-out waveguides, instead of forcing all optical power to stay in the main network and degrade the final classification result. The paper analytically compared the diamond mesh with the Reck mesh for different ONN sizes and showed that the diamond structure is more tolerant to two important practical nonidealities: MZI insertion loss and phase errors. In this sense, the contribution of the paper is not just proposing another interferometer layout, but showing that mesh topology itself can improve robustness and classification accuracy for programmable MZI-based ONNs.

1.1.10 Optical coherent dot-product chip for sophisticated deep learning regression

Authors: Shaofu Xu et al.
Journal: Light: Science & Applications 10, 221 (2021)
DOI: 10.1038/s41377-021-00666-8

Innovation, working principle, and use. This work focused on the most basic neural-network primitive: the dot product. The processor uses coherent interference, so each input channel carries an optical field with both amplitude and phase, and the relative phase determines whether a weighted contribution adds constructively or destructively at the output. That is important because many machine-learning workloads require signed weights, not only nonnegative intensities. The chip therefore implements multiplication by controlling the complex transmission of each path and implements summation by interfering the weighted fields in a shared output. The authors emphasized regression-oriented tasks, which are harder than simple winner-take-all classification because the processor must preserve numerical relationships in the output rather than only identify the brightest class. The specific innovation of the paper is to show that coherent silicon photonics can act as a compact signed dot-product engine suitable for more numerically demanding deep-learning workloads.

1.1.11 Nonlinear germanium-silicon photodiode for activation and monitoring in photonic neuromorphic networks

Authors: Yang Shi et al.
Journal: Nature Communications 13, 6048 (2022)
DOI: 10.1038/s41467-022-33877-7

Innovation, working principle, and use. This paper used the photodiode as more than a passive receiver. The device is a germanium-silicon photodiode integrated in a silicon photonic

platform, so it can be placed directly at the output of an on-chip optical linear layer. Its photocurrent is intentionally made nonlinear with respect to the incident optical power, which means the same device can convert a linear optical sum into a nonlinear electrical output suitable for neuron activation. At the same time, because it is a detector, it also exposes internal optical states that can be used for monitoring, calibration, or training supervision. This dual role is the key innovation: instead of inserting one component for activation and another for observability, the paper combines both functions in a single silicon-compatible device. For practical photonic neural networks, that integration matters because monitoring overhead is one of the main hidden costs of large optical circuits.

1.1.12 Single-chip photonic deep neural network with forward-only training

Authors: Saumil Bandyopadhyay et al.
Journal: Nature Photonics 18(12), 1335–1343 (2024)
DOI: 10.1038/s41566-024-01567-z

Innovation, working principle, and use. This paper moved beyond the usual "offline train, then load phases" workflow and demonstrated a single chip that can be trained by interacting with measured forward passes. The chip integrates optical linear layers with on-chip nonlinear processing so that a complete deep network, not just an isolated matrix operator, is present on one photonic platform. The distinctive idea is forward-only training: instead of explicitly transporting exact gradients backward through every nonideal optical component, the learning algorithm updates the chip parameters from forward measurements and hardware-aware feedback. That makes the method more tolerant of model mismatch, fabrication error, and drift, because the update rule is anchored to what the real chip actually does rather than to a perfect simulation. For a beginner, the important message is that the authors are not claiming optics somehow removes training; they are showing a practical way to train a photonic network when standard backpropagation is cumbersome on analog hardware. The paper is therefore a major step toward trainable monolithic photonic deep-learning chips.

1.1.13 I/O-efficient iterative matrix inversion with photonic integrated circuits

Authors: Minjia Chen et al.
Journal: Nature Communications 15, 5926 (2024)
DOI: 10.1038/s41467-024-50302-3

Innovation, working principle, and use. The photonic circuit is used as a fast matrix-vector engine inside an iterative solver, so the paper attacks not only arithmetic speed but also the data-movement bottleneck that usually hides optical speedups. This work presented an iterative matrix-inversion strategy implemented with photonic integrated circuits while explicitly attacking the usual I/O bottleneck. Matrix inversion is not itself a standard neural-network layer, but it is a representative linear-algebra kernel that reveals how photonic processors can reuse optical information to reduce data movement overhead. The paper therefore matters for photonic neural computing because it addresses a systems issue that often erodes the theoretical benefit of optical acceleration. It also shows that optical hardware design must consider dataflow and reuse, not only raw arithmetic throughput.

1.1.14 Photonic Neural Network Fabricated on Thin Film Lithium Niobate for High-Fidelity and Power-Efficient Matrix Computation

Authors: Yong Zheng et al.
Journal: Laser & Photonics Reviews 18(10), 2400565 (2024)
DOI: 10.1002/lpor.202400565

Innovation, working principle, and use. Here the main device advantage comes from thin-film lithium niobate: the Pockels effect gives fast, low-drift phase tuning, so a mesh can be programmed with less static power than thermo-optic silicon counterparts. This paper reported a photonic neural network fabricated on thin-film lithium niobate and emphasized high-fidelity, high-speed, and power-efficient matrix computation. The architecture used an electro-optically tunable MZI-style mesh on a TFLN platform, showing that lithium niobate can support accurate programmable linear computation for neural tasks while benefiting from fast electro-optic control. The work is important because it extends photonic neural-network hardware beyond mainstream silicon platforms and demonstrates in situ training-oriented functionality on a fast material system. It helps position TFLN as a strong candidate for future photonic AI chips.

1.1.15 Silicon photonic convolution operator exploiting on-chip nonlinear activation function

Authors: Zilong Tao et al.
Journal: Optics Letters 50(2), 582–585 (2025)
DOI: 10.1364/OL.543024

Innovation. The key step forward is that the chip does not add a separate nonlinear-activation block after the convolution block. Instead, it uses the nonlinear modulation physics of an on-chip electro-optic IQ modulator itself as the activation function, so convolution and activation happen in one integrated optical path. The paper presents this as the first implementation of that idea on a silicon photonic convolution chip.

Working principle. The chip receives two vectors: one carries the input data and the other carries the convolution kernel. Both are encoded as RF signals in an OFDM-like format and then loaded onto optical carriers by two IQ modulators. A 90° optical hybrid and balanced photodetectors recover the convolution through beat-frequency mixing, so the linear part of the chip performs the same role as a standard optical convolution operator. The new part appears when the RF drive on the data arm is made large. At low RF amplitude, the IQ modulator behaves almost linearly. At high RF amplitude, however, the modulator generates many higher-order sidebands through harmonic generation and self-mixing, with amplitudes described by Bessel functions. In effect, the modulator no longer outputs the original data vector A, but a nonlinear mapping $g(A)$. The chip therefore computes $g(A) * B$ rather than $A * B$: the input is nonlinearly transformed inside the modulator, and the transformed signal is convolved with the kernel on the same chip. This is how the activation is merged into the optical front end without adding another optical-electrical-optical module.

How to use it. To use this design, encode the input data as an RF waveform on the first IQ modulator, encode the kernel on the second arm, and deliberately drive the first modulator into its nonlinear regime so it generates the desired activation mapping. The resulting beat signal from the hybrid and balanced photodetectors is the convolution of the activated input with the kernel. The authors then used this operator inside a CNN for the RadioML 2016.10b dataset and reached 92.5% classification accuracy, which was 27 percentage points higher than the same model without the nonlinear activation.

1.1.16 Reconfigurable photonic neural networks: distribution-aligned calibration and dynamic resource allocation for high-performance optical computing accelerators

Authors: Songcheng Zhou et al.
Journal: Optics Express 33(21), 45413–45435 (2025)
DOI: 10.1364/OE.574781

Innovation, working principle, and use. The paper treats calibration as a workload-aware problem: the hardware should be tuned around the statistics of the actual input distribution instead of around an abstract worst-case transfer matrix. This paper tackled the deployment problem of reconfigurable photonic neural networks by introducing distribution-aligned calibration and dynamic resource allocation. The main idea is that calibration should not be done in a task-agnostic way; instead, hardware tuning and resource assignment should reflect the actual data distribution and model needs. This is an important practical contribution because photonic accelerators often lose accuracy when calibration and workload assumptions diverge. The work therefore improves the realism of running optical neural networks under changing operating conditions.

1.1.17 Cross-domain knowledge distillation for integrated optical neural networks: toward robust and compact learning with pseudo-zero-shot capability

Authors: Zongyu Lu et al.
Journal: Optics Express 33(22), 45483–45498 (2025)
DOI: 10.1364/OE.570333

Innovation, working principle, and use. The key idea is algorithmic rather than purely photonic: a stronger teacher model transfers task structure into a smaller optical student that is easier to fit within the precision and scale limits of integrated hardware. This paper proposed a cross-domain knowledge-distillation framework for integrated optical neural networks. Instead of only redesigning the photonic hardware, the authors used distillation to transfer useful structure from stronger models into a compact optical network that is easier to implement under hardware constraints. The contribution is important because algorithmic compression and robustness methods are increasingly necessary if integrated optical neural networks are to stay accurate at modest chip scale. It also suggests that pseudo-zero-shot style generalization may help photonic models cope with limited on-chip capacity.

1.1.18 Efficient hardware error correction with hybrid on-offline configuration algorithm for optical processor

Authors: Zichao Zhao et al.
Journal: Communications Physics 8, 323 (2025)
DOI: 10.1038/s42005-025-02247-2

Innovation. The new idea is not a new optical primitive, but a new way to program a nonideal optical processor quickly and reliably. The authors combine a rough offline preset with a short online correction stage, so the chip starts near the right solution and does not waste many iterations searching from a random point.

Working principle. The offline stage first maps a trained matrix onto the physical mesh. For a neural-network layer, the target matrix W is decomposed as $W = U\Sigma V^\dagger$, and these factors are encoded on Clements or FFT-style MZI meshes. Instead of performing a slow, high-precision heater calibration, the chip uses near-field infrared monitoring ports and only a small number of voltage samples to build an approximate phase-voltage map. That gives initial phase settings for the

heaters. The online stage then corrects what the offline stage cannot capture: splitter imbalance, path-dependent loss, thermal crosstalk, and other fabrication errors. The authors use a perturbed optimization iteration method. During each iteration, they slightly perturb a phase, reconstruct the actual matrix from measured outputs, compare it with the target matrix, and update the phases in the direction that reduces the error. A key detail is the loss function: both the target matrix and the measured matrix are normalized by the Frobenius norm, so the algorithm can correct the matrix shape even when device loss changes the absolute optical power.

How to use it. To use this method, first decompose the target neural-network matrix and load the resulting rough phase values onto the chip. Then send basis inputs through the processor, reconstruct the implemented matrix from the monitored outputs, run the perturb-and-update loop for a few rounds, and stop when the fidelity saturates. In the paper, 4×4 Clements and FFT meshes reached matrix fidelities above 0.98 in fewer than 10 iterations, and the same corrected meshes were then used for Iris and Vowel classification tasks.

1.1.19 Data-driven programming of feedforward photonic integrated circuits

Authors: Gabriele Cavicchioli et al.
Journal: APL Photonics 11(1), 010801 (2026)
DOI: 10.1063/5.0293082

Innovation. This paper treats photonic programming as a learned inverse problem. Instead of trying to write down a perfect physical model of a real MZI mesh and then inverting that model, it trains a neural network directly on measured chip data, so the controller learns how the fabricated chip actually behaves.

Working principle. A feedforward MZI mesh is described by a complex transfer matrix H, but the control variables are the electrical powers applied to the heaters. In a real chip, that mapping is hard to invert because the heaters interact through thermal crosstalk, so the phase shifts satisfy $\theta = KP$, and the overall relation between heater powers P and optical matrix H is nonlinear and periodic. The authors therefore train a fully connected ANN with five dense layers to approximate the inverse map $H \mapsto P$. The input to the ANN is the desired complex transfer matrix, written as its real and imaginary parts and flattened into a real vector; the output is one heater-power value per phase shifter. To train the model, they generate many examples by programming heater powers on the mesh and measuring the corresponding transfer matrices. The training set has two parts: a large random dataset that covers the state space broadly, and a sweep dataset that varies one actuator at a time so the network learns the sinusoidal response of each phase shifter. In the experimental 3×3 triangular silicon mesh, the controller is trained on 10^6 in-situ records and then predicts heater powers that already compensate nonidealities such as thermal crosstalk.

How to use it. In practice, the workflow is: measure many input-output examples from the real chip, train the ANN once, and then use the trained ANN like a compiler. Whenever a user wants a new transfer matrix, they feed that target matrix into the ANN, read out the predicted heater powers, and apply those powers directly to the chip. No explicit crosstalk matrix extraction is needed. On the experimental 3×3 mesh, the average matrix-element squared error was around 2%, and badly programmed cases were statistically rare, showing that the method can program a real feedforward mesh from standard transmission measurements alone.

1.1.20 Inverse-designed nanophotonic neural network accelerators for ultra-compact optical computing

Authors: Joel Sved et al.
Journal: Nature Communications 17, 1059 (2026)
DOI: 10.1038/s41467-026-68648-1

Innovation, working principle, and use. This paper replaces the usual "mesh of discrete interferometers" view of photonic computing with a much denser inverse-designed optical region. Instead of programming a few phase shifters, the authors treat each subwavelength voxel in a tiny silicon nanophotonic block as a trainable design variable and optimize the whole electromagnetic structure with three-dimensional finite-difference time-domain simulations. The device then performs the learned optical transform through coherent scattering and interference inside an area as small as tens of micrometers. The key point is that the computation is no longer distributed over many large building blocks such as MZIs; it is compressed into a task-optimized nanophotonic pattern with extremely high parameter density. Experimentally, the paper demonstrated ultra-compact on-chip classifiers for MNIST and MedNIST, which showed that inverse design can produce useful neural-network accelerators in footprints far smaller than conventional programmable meshes. The specific innovation is therefore ultra-compact, application-specific optical computing obtained by letting Maxwell-equation optimization design the entire physical structure directly.

1.1.21 LightIN: a versatile silicon-integrated photonic field programmable gate array with an intelligent configuration framework for next-generation AI clusters

Authors: Ying Zhu et al.
Journal: Light: Science & Applications 15, 165 (2026)
DOI: 10.1038/s41377-026-02209-5

Innovation, working principle, and use. This paper presents a photonic processor as something closer to an FPGA than to a single fixed accelerator. The hardware is a silicon photonic square mesh built from many programmable unit cells, each based on an MZI with thermo-optic control, so the same chip can be compiled into different linear transformations and networking functions. The important novelty is the full configuration stack around the chip: the authors developed a testing, compilation, and adjustment framework that programs the mesh without requiring dedicated on-chip monitor photodetectors everywhere. Using this combination of optical fabric and control software, they demonstrated matrix multiplication, neural-network inference, wavelength locking, optical switching, and even security-oriented photonic physical unclonable functions. So the innovation is not only one fast matrix engine; it is a versatile silicon photonic field-programmable platform plus an intelligent configuration workflow, aimed at giving AI clusters one reusable optical substrate for computing, interconnect, and signal processing.

1.2 MZI-based incoherent PNN

These papers keep programmable interferometric weighting but avoid relying on one large phase-stable coherent interference pattern. In most cases, information is distributed across wavelength, time, or space channels, the chip applies channel-by-channel weights, and photodetectors add the optical powers at the end. I therefore focus on the specific encoding method, weighting path, and readout strategy used by each design.

This subsection will cover incoherent MZI-based photonic neural-network approaches, with emphasis on implementation strategies and trade-offs.

1.2.1 Photonic Perceptron Based on a Kerr Microcomb for High-Speed, Scalable, Optical Neural Networks

Authors: Xingyuan Xu, Mengxi Tan, Bill Corcoran, Jiayang Wu, Thach G. Nguyen, Andreas Boes, Sai T. Chu, Brent E. Little, Roberto Morandotti, Arnan Mitchell, Damien G. Hicks, and David J. Moss
Journal: Laser & Photonics Reviews 14, 2000070 (2020)
DOI: 10.1002/lpor.202000070

Innovation, working principle, and use. This paper proposed a photonic neural-network architecture based on an *integrated Kerr microcomb*, with the goal of overcoming the scalability limits of discrete-laser or purely spatial ONN schemes. The key idea was to use a single microcomb as a massively parallel multi-wavelength source, where each comb line serves as one synapse. In the experimental demonstration, the authors built the basic ONN building block—a *single-layer, single-neuron perceptron*—using *49 wavelength channels*, enabled by a *49 GHz* microcomb spacing. The input data were first converted into a 49-element vector, then serialized in time and multicast onto all comb lines. A programmable spectral shaper encoded the trained synaptic weights onto the optical power of each wavelength, so that wavelength multiplexing performed the weighting while time multiplexing carried the input symbols. After modulation, a dispersive delay element progressively shifted the weighted wavelength channels in time so that the terms $w(1)x(1), w(2)x(2), \ldots, w(49)x(49)$ aligned into the same time slot and were summed by photodetection, thereby realizing the vector dot product. The nonlinear activation was applied afterward in the electrical domain. Using this architecture, the paper reported a *single-unit throughput of 11.9 GigaFLOPS*, equivalent to *95.2 Gbps at 8 bits per FLOP*. The perceptron was tested on two binary classification tasks: handwritten-digit recognition, where it achieved about *93.75%* experimental accuracy, and benign-versus-malignant cancer-cell classification, where it achieved about *86.67%*. The paper further proposed how the same microcomb-based principle could be extended to multi-layer ONNs by jointly using wavelength, time, and spatial multiplexing, so its main contribution was to show that microcombs can act as a highly scalable synaptic source for high-speed optical neural computation rather than just as a convenient WDM light source.

1.2.2 Integrated Photonic Neural Network Based on Silicon Metalines

Authors: Sanaz Zarei, Mahmood-Reza Marzban, and Amin Khavasi
Journal: Optics Express 28(24), 36668–36684 (2020)
DOI: 10.1364/OE.404386

Innovation, working principle, and use. This paper proposed an *integrated diffractive-style photonic neural network* built from cascaded one-dimensional silicon metasurfaces, which the authors called *metalines*. Instead of following the more common MZI-mesh or microring-weight-bank route, the work used a sequence of passive high-contrast transmitarray metasurfaces patterned on a silicon-on-insulator platform. Each metaline was designed to apply a trained spatial transformation to the optical field, while propagation between successive metalines provided the mode mixing needed for network operation. In this way, the overall device implemented the neural-network matrix operations through wave propagation and engineered phase control, rather than through many separately tuned interferometers or resonators. A key point of the paper is that the whole architecture was intended to be *fully passive, fully optical, compact, and low-loss*, so the computation in principle occurs at the speed of light with very low energy consumption once the structure is fabricated. The paper further benchmarked the concept on handwritten-digit classification from the MNIST dataset at a wavelength of 1.55 μm, arguing that the silicon-metaline platform can perform

complex machine-learning tasks while remaining much more compact than bulk free-space diffractive neural networks. So the main contribution was to bring the diffractive-neural-network idea into a much more integrated on-chip form by replacing large free-space phase masks with cascaded silicon metalines.

1.2.3 11 TOPS photonic convolutional accelerator for optical neural networks

Authors: Xingyuan Xu et al.
Journal: Nature 589(7840), 44–51 (2021)
DOI: 10.1038/s41586-020-03063-0

Innovation, working principle, and use. This is one of the landmark papers showing that photonic hardware can execute convolution, not only dense matrix multiplication, at very high throughput. The architecture uses many wavelength channels generated from a microcomb so that multiple input data streams and kernel weights can be carried in parallel on one optical link. Modulators write image data onto the comb lines, weighting elements implement the kernel coefficients, and photodetection performs the accumulation step needed for convolution. By carefully arranging wavelength, time, and reuse of the optical hardware, the same photonic core can slide a kernel across the input much more efficiently than a naive fully connected mapping. That is why the work matters: convolution is the dominant operation in many vision networks, and the paper showed that integrated photonics can be organized around this structure directly instead of treating every CNN as a generic matrix multiplier. The reported 11 TOPS figure made it a flagship demonstration of photonic convolution acceleration.

1.2.4 An on-chip photonic deep neural network for image classification

Authors: Farshid Ashtiani et al.
Journal: Nature 606(7914), 501–506 (2022)
DOI: 10.1038/s41586-022-04714-0

Innovation, working principle, and use. This paper is important because it demonstrated a real on-chip image classifier rather than only a photonic linear-algebra kernel. The network combines optical weighting, summation, and nonlinear processing in one integrated photonic-electronic workflow, so the chip executes the layers needed for classification instead of serving merely as a front-end operator. The optical part performs the parallel weighted sums, while the nonlinear and control stages complete the neuron function and enable multiple layers to be cascaded. For a reader new to the area, the key advance is system completeness: the authors showed that a fabricated chip can accept image-derived inputs, propagate them through several neural-network operations, and produce class outputs on hardware. That made the work a reference point for later efforts aiming at end-to-end photonic inference, because it moved the field beyond isolated demonstrations of one matrix multiply.

1.2.5 InP photonic integrated multi-layer neural networks: Architecture and performance analysis

Authors: Bin Shi et al.
Journal: APL Photonics 7(1), 010801 (2022)
DOI: 10.1063/5.0066350

Innovation, working principle, and use. The importance of this paper is architectural: it studies how an indium-phosphide platform can co-integrate active devices such as lasers, modulators,

and detectors to build multi-layer photonic networks. This paper analyzed a multi-layer neural-network architecture built on an InP photonic integrated platform, emphasizing the co-integration of optical synapses and nonlinear neuron functions. Using the capabilities of InP, the authors discussed how active devices such as semiconductor optical amplifiers can support multi-layer all-optical processing more naturally than a passive silicon-only platform. The work is important because it expands the material-platform discussion in photonic neural networks and shows that active InP integration may help address gain and nonlinearity bottlenecks.

1.2.6 Symmetric silicon microring resonator optical crossbar array for accelerated inference and training in deep learning

Authors: Rui Tang et al.
Journal: Photonics Research 12(8), 1681–1688 (2024)
DOI: 10.1364/PRJ.520518

Innovation, working principle, and use. The optical crossbar uses a symmetric ring layout to improve how input channels, weight channels, and gradient paths are matched, making the same fabric more suitable for both inference and training-related updates. This paper introduced a symmetric silicon microring-resonator optical crossbar array aimed at both inference and training acceleration. The symmetry of the hardware makes forward and transpose-like operations easier to reuse, which is attractive for backpropagation-related computation on chip. The work is important because it tries to bridge the gap between photonic inference engines and photonic training hardware, not only static deployment. It also demonstrates how circuit symmetry can become a useful architectural principle for optical deep learning hardware.

1.2.7 Photonic edge intelligence chip for multi-modal sensing, inference and learning

Authors: Shiji Zhang et al.
Journal: Nature Communications 16(1), 10136 (2025)
DOI: 10.1038/s41467-025-65151-x

Innovation, working principle, and use. This paper is notable because it merges sensing and neural inference on the same photonic chip. Instead of first converting every analog signal into digital data and then sending it to a separate processor, the authors encode several kinds of raw information—images, spectra, and radio-frequency signals—into a broad optical spectrum carried by a single fiber. Once the signal reaches the chip, an arrayed waveguide grating separates the spectrum and simultaneously performs a convolution-like weighting operation, so the same optical front end acts as both a sensor interface and a compute block. A nonlinear activation layer and a fully connected layer then finish the neural-network inference on chip. The specific innovation is this unified "sense and compute" pipeline: multimodal analog data are fused into light, processed by spectral photonic hardware, and classified within about a nanosecond-scale response time, which is exactly the kind of architecture edge intelligence needs.

1.2.8 Integrated lithium niobate photonic computing circuit based on efficient and high-speed electro-optic conversion

Authors: Yaowen Hu et al.
Journal: Nature Communications 16(1), 8178 (2025)
DOI: 10.1038/s41467-025-62635-8

Innovation, working principle, and use. This paper addressed one of the least glamorous but most important bottlenecks in photonic computing: electro-optic conversion. Even when the

optical core is fast, the overall system is limited if writing data into light is slow or power-hungry. The authors therefore built the computing engine on thin-film lithium niobate, whose strong Pockels effect enables very efficient, high-speed modulation. The resulting circuit combines more than one hundred TFLN components so that electrical data can be converted into weighted optical signals at high speed, processed on chip, and detected with very low energy per operation. They also pushed the integration level by combining the computing circuit with an integrated laser source and high-speed photodiode technologies. The specific innovation is thus a system-level TFLN photonic computing circuit in which the electro-optic interface itself is part of the advantage, not merely a peripheral component.

1.2.9 Scaling up for end-to-end on-chip photonic neural network inference

Authors: Bo Wu et al.
Journal: Light: Science & Applications 14, 328 (2025)
DOI: 10.1038/s41377-025-02029-z

Innovation, working principle, and use. This work directly targeted the two scaling problems of earlier end-to-end optical neural networks: they were usually too shallow because nonlinear activations did not cascade well, and they usually accepted only small inputs because the optical matrix size was limited. The authors proposed what they call a partially coherent deep optical neural network, or PDONN. Its first key component is an on-chip opto-electro-opto activation that provides positive net gain, so one layer can drive the next instead of dying out after repeated propagation. Its second key component is the use of convolutional layers to reduce dimensionality early, which lets the system handle larger inputs without requiring an unmanageably large fully connected optical matrix. The use of partially coherent light is also deliberate: it relaxes the need for narrow-linewidth lasers and full coherent detection, which makes the system easier to scale in practice. The paper therefore stands out as a carefully engineered route to deeper and larger end-to-end on-chip optical inference, not just a one-off improvement in one device.

1.2.10 Time-space-wavelength multiplexed photonic tensor core for deep-learning acceleration

Authors: A. Prapas et al.
Journal: Optics Express 33(17), 36960–36972 (2025)
DOI: 10.1364/OE.564666

Innovation, working principle, and use. The tensor core exploits several orthogonal multiplexing dimensions at once, so one physical device reuses the same modulators and waveguides across many tensor elements instead of dedicating hardware to one scalar multiply. This paper introduced a photonic tensor core that combines time, space, and wavelength multiplexing to boost effective parallelism. The architecture tries to extract more computational density from integrated photonics by stacking multiple multiplexing dimensions rather than enlarging a single static optical core. This is important for deep-learning acceleration because optical hardware often reaches footprint and routing limits before it reaches the desired throughput. Multidomain multiplexing therefore provides a practical route to scaling performance on chip.

1.2.11 High-integrated photonic tensor core utilizing high-dimensional lightwave and microwave multidomain multiplexing

Authors: Xiangyan Meng et al.
Journal: Light: Science & Applications 14, 27 (2025)

DOI: 10.1038/s41377-024-01706-9

Innovation, working principle, and use. The architecture mixes optical and microwave domains together with multiple optical dimensions, pushing photonic tensor processing beyond simple wavelength-only parallelism. This work developed a highly integrated photonic tensor core using high-dimensional lightwave and microwave multidomain multiplexing. By jointly exploiting optical and microwave domains, the design targets very high throughput without linearly increasing the number of static photonic components. The paper is relevant because tensor-core style accelerators are one of the most promising ways to map neural-network arithmetic onto photonic hardware. It also illustrates how cross-domain multiplexing can become a serious design strategy for next-generation photonic AI accelerators.

1.2.12 High-speed electro-optic engine for a quantized photonic neural network

Authors: Fengkai Han et al.
Journal: Optica 13(2), 236–246 (2026)
DOI: 10.1364/OPTICA.584297

Innovation. This paper combines a fast thin-film-lithium-niobate hardware platform with a quantized neural-network algorithm. The important hardware idea is the input-weight parallel encoding scheme: instead of using one device to encode the input and another device to encode the weight, a single electro-optic modulator performs the multiplication while a slow bias controller sets the weight. The important algorithm idea is to use low-bit inputs and low-bit weights so the optical hardware does not need full analog precision at 100 Gbaud.

Working principle. The electro-optic engine contains eight LNOI channels with electro-optic bandwidth above 67 GHz. In the weight-stationary version of the encoding scheme, the high-speed electrical input stream is sent to the electro-optic modulator, while the weight is represented by the modulator bias point set by a thermal controller. Different bias points give different output slopes, which correspond to weights such as $+1$, $+1/2$, 0, $-1/2$, and -1. In other words, the same input waveform is multiplied by different fixed weights simply by sending it through channels biased at different operating points. The optical outputs are then detected and used as the multiplied results. Quantization is crucial here: when the authors compare high-precision encoding with quantized encoding at high speed, the quantized case is more tolerant to waveform distortion, so the hardware can exploit the full modulation bandwidth more effectively.

How to use it. The practical workflow is to train a quantized neural network in software, set one bias value per channel to represent the desired fixed weights, flatten the input data into a time sequence, and then stream that quantized sequence through the LNOI engine at 100 Gbaud. The detected optical outputs become the first-layer multiply results and are passed to the remaining layers. Using this method, the paper experimentally implemented the first convolutional layer of a quantized CNN for MNIST and reached 97% accuracy in the fully quantized 100 Gbaud case, with more than 95% model compression. The same hardware was also used for a quantized GCN on the Cora dataset and reached 80% node-classification accuracy.

1.3 MRR-based PNN

The common picture in this subsection is broadcast-and-weight computing. Different wavelengths carry different channels, each microring selects or scales one wavelength, and the detector sums the weighted powers. Rather than repeating that background in every paper, the entries below emphasize what each design adds: better neurons, better control, better multiplexing, or better system integration.

1.3.1 Silicon Photonic Modulator Neuron

Authors: Alexander N. Tait, Thomas Ferreira de Lima, Mitchell A. Nahmias, Heidi B. Miller, Hsuan-Tung Peng, Bhavin J. Shastri, and Paul R. Prucnal
Journal: Physical Review Applied 11, 064043 (2019)
DOI: 10.1103/PhysRevApplied.11.064043

Innovation, working principle, and use. Here the modulator is used as a neuron primitive: optical inputs are weighted and summed, converted to an electrical drive, and then re-encoded optically so one stage can fan out to the next. This paper followed a different route from coherent MZI-mesh ONNs: instead of implementing matrix multiplication by interference, it demonstrated a *network-compatible photonic neuron* on a silicon photonics platform. The core device is an optical-electrical-optical neuron formed by a balanced photodetector directly connected to a microring modulator. Multiple wavelength-multiplexed optical inputs are first weighted and summed at the photodetector, converted into an electrical signal, and then used to modulate a new optical carrier through the microring. In this way, the device realizes fan-in, nonlinear transfer, and wavelength conversion in one unit, while also supporting inhibitory operation through balanced detection. The paper further used an autapse experiment and observed bistability as evidence of indefinite cascadability, arguing that this device provides a key neuron element for fully integrated broadcast-and-weight photonic neural networks.

1.3.2 Feedback Control for Microring Weight Banks

Authors: Alexander N. Tait, Hasitha Jayatilleka, Thomas Ferreira de Lima, Philip Y. Ma, Mitchell A. Nahmias, Bhavin J. Shastri, Sudip Shekhar, Lukas Chrostowski, and Paul R. Prucnal
Journal: Optics Express 26(20), 26422–26443 (2018)
DOI: 10.1364/OE.26.026422

Innovation, working principle, and use. This paper addressed a very practical bottleneck in microring-based photonic neural networks: not how to design a weight bank in principle, but how to *control it reliably* under fabrication error and thermal drift. The target hardware is a broadcast-and-weight microring bank, where each add-drop microring resonator sets the weight on one WDM channel. Instead of relying only on pre-measured calibration curves and open-loop feedforward tuning, the authors introduced a *feedback control* method based on in-ring N-doped photoconductive heaters. These heaters serve two roles at once: they thermally tune the ring resonance, and they also sense the ring state electrically, because their resistance changes with temperature and with photoabsorption-induced carrier generation from the circulating optical field. The key idea is therefore to estimate the ring transmission weight from electrical observables, mainly heater current and measured voltage, rather than continuously measuring the optical spectrum during operation. The paper developed a simplified control model in which commanded weight is translated into a target electrical-response variable, and the heater bias is then adjusted by feedback until the estimated state reaches the target. In this way, the control loop does not need to solve the full coupled thermal-optical physical model online, even though the real device is affected by self-heating, optical absorption, wavelength detuning, and shared electrical parasitics. The authors experimentally characterized this behavior for different carrier power and wavelength offsets, and demonstrated continuous multi-channel weight control with effective 5.1-bit accuracy on 2 Gbps signals. They also showed that even without direct transmission measurements, the feedback estimate alone could still provide 4.2-bit accuracy, which is important because it makes weight control much more scalable for larger analog photonic networks.

1.3.3 Demonstration of Scalable Microring Weight Bank Control for Large-Scale Photonic Integrated Circuits

Authors: Chaoran Huang, Simon Bilodeau, Thomas Ferreira de Lima, Alexander N. Tait, Philip Y. Ma, Eric C. Blow, Aashu Jha, Hsuan-Tung Peng, Bhavin J. Shastri, and Paul R. Prucnal
Journal: APL Photonics 5(4), 040803 (2020)
DOI: 10.1063/1.5144121

Innovation, working principle, and use. This paper focused on a key practical bottleneck in microring-based photonic neural and analog processors: how to *control a microring weight bank accurately at scale*. Rather than proposing a new computing architecture, it addressed the calibration and tuning problem of multi-channel microring resonator (MRR) weight banks for large-scale photonic integrated circuits. The main contribution was a *simple and scalable calibration/control procedure* that enables continuous multi-channel tuning of ring weights while avoiding the need for direct access to the output of the weight bank and avoiding extra optical or electrical I/Os dedicated only to calibration. In this sense, the paper moved the field from small proof-of-concept weight tuning toward a more system-compatible control strategy that could be embedded inside larger photonic circuits. Experimentally, the authors demonstrated continuous control of a four-channel weight bank and reported *record-high accuracy and precision up to 7 bits*, together with negligible inter-channel crosstalk when other channels were tuned across their full transmission range. They also studied the usable dynamic range of the method and showed that an input optical power as low as -12 dBm was sufficient to achieve 5-bit weight resolution, which is already relevant for neuromorphic systems. In addition, similar accuracy was maintained for different initial frequency offsets spanning a 100 GHz WDM channel spacing, suggesting that the scheme is tolerant to practical wavelength offsets and compatible with cost-effective WDM laser sources. Overall, the paper's importance is that it made microring weight control look much more *scalable and deployment-oriented*, not just physically possible in a small laboratory setup.

1.3.4 Silicon photonic architecture for training deep neural networks with direct feedback alignment

Authors: Matthew J. Filipovich et al.
Journal: Optica 9(12), 1323–1332 (2022)
DOI: 10.1364/OPTICA.475493

Innovation, working principle, and use. Direct feedback alignment replaces the exact transpose of the forward weights with a fixed random feedback path, which is easier to implement optically than strict backpropagation but can still drive learning. This work proposed a silicon photonic deep-learning architecture trained with direct feedback alignment rather than standard backpropagation. The idea is attractive for hardware because direct feedback alignment relaxes some symmetry and routing requirements that make conventional photonic backpropagation difficult to realize on chip. By mapping the training method onto an integrated photonic architecture, the paper showed that alternative learning rules may be especially useful for analog optical hardware where exact gradient transport is costly.

1.3.5 A silicon photonic-electronic neural network for fibre nonlinearity compensation

Authors: Chaoran Huang et al.
Journal: Nature Electronics 4, 837–844 (2021)
DOI: 10.1038/s41928-021-00661-2

Innovation, working principle, and use. The photonic front end accelerates the linear algebra while electronics handle the remaining nonlinear processing needed to undo distortion in optical-fiber communication links. This paper presented a silicon photonic–electronic neural network tailored to fibre nonlinearity compensation in optical communication systems. The architecture combined photonic matrix operations with electronic nonlinear processing to implement neural equalization at high data rates. Rather than targeting only image classification, the work addressed a concrete communications problem and showed that integrated photonic neural hardware can solve physically meaningful regression and compensation tasks with performance competitive with software-based neural networks.

1.3.6 Microcomb-based integrated photonic processing unit

Authors: Bowen Bai et al.
Journal: Nature Communications 14, 66 (2023)
DOI: 10.1038/s41467-022-35506-9

Innovation, working principle, and use. This paper is a representative example of the microcomb route to photonic AI hardware. A single microcomb generates many evenly spaced wavelengths, which immediately solves a major scalability problem: one source can provide dozens of parallel optical carriers instead of requiring many discrete lasers. Input data are written onto those comb lines with modulators, wavelength-selective weights are applied downstream, and photodetectors sum the optical powers to realize many multiply–accumulate operations at once. Because the comb lines are naturally phase-locked and densely spaced, the hardware becomes very compact and highly parallel. The innovation of the paper is to package source generation, multiplexing, weighting, and readout into one integrated photonic processing unit rather than demonstrating the microcomb only as a light source. That made the microcomb a central computing resource inside the processor itself.

1.3.7 Integrated WDM-compatible optical mode division multiplexing neural network accelerator

Authors: Ruoyu Yin et al.
Journal: Optica 10(12), 1709–1718 (2023)
DOI: 10.1364/OPTICA.500523

Innovation, working principle, and use. The paper combines wavelength-division multiplexing with mode-division multiplexing so more independent channels can share the same chip area and carry a larger matrix operation in parallel. This paper proposed an integrated neural-network accelerator that combines wavelength-division multiplexing with mode-division multiplexing. By adding spatial mode channels to wavelength channels, the architecture increases parallelism without depending only on more lasers or more wavelength resources. The paper is important because it tackles an overlooked scaling issue in photonic neural networks: parallelism is often limited not just by component count, but also by how efficiently optical channels are multiplexed.

1.3.8 Control-free and efficient integrated photonic neural networks via hardware-aware training and pruning

Authors: Tengji Xu et al.
Journal: Optica 11(8), 1039–1049 (2024)
DOI: 10.1364/OPTICA.523225

Innovation, working principle, and use. Rather than eliminating device errors physically, the paper uses training and pruning rules that adapt the model to the measured hardware so the final network needs much less active post-fabrication control. This paper proposed a hardware-aware training and pruning strategy that drives integrated photonic neural networks toward operating regions that are naturally more robust and energy efficient. Instead of relying on heavy active control during operation, the method shapes the trained model so that it can tolerate hardware imperfections with much less stabilization overhead. This is a major practical contribution because complex control loops are one of the main reasons many promising PIC neural architectures remain hard to scale.

1.3.9 120 GOPS Photonic tensor core in thin-film lithium niobate for inference and in situ training

Authors: Zhongjin Lin et al.
Journal: Nature Communications 15, 9081 (2024)
DOI: 10.1038/s41467-024-53261-x

Innovation, working principle, and use. This paper transferred the popular tensor-core idea into a thin-film lithium-niobate photonic platform. The hardware uses fast electro-optic modulation in TFLN to encode tensor elements into optical channels, and the optical network then executes many multiply–accumulate operations in parallel. What makes the paper more than a simple forward accelerator is that the same platform is also used for in situ training-related operations, so the authors are thinking about how the tensor core can support both inference and parameter updates. For beginners, the key point is that a tensor core is a specialized matrix engine optimized for the block operations common in modern AI workloads. This paper showed that lithium niobate can host such a block-structured photonic engine at high speed, making TFLN not only a fast modulator platform but a serious candidate for AI-oriented optical compute blocks.

1.3.10 Monolithically integrated asynchronous optical recurrent accelerator

Authors: Bo Wu et al.
Journal: eLight 5, 7 (2025)
DOI: 10.1186/s43593-025-00084-y

Innovation, working principle, and use. This paper is important because recurrent optical computing becomes especially difficult when many parallel signals must stay synchronized at high speed. The authors avoided that bottleneck by proposing an asynchronous recurrent architecture based on wavelength encoding. Instead of representing the recurrent state by tightly clocked electrical timing, the next state is relayed through an ordered set of wavelength channels, so the recurrence follows spectral causality rather than strict electronic synchronization. They then built this idea monolithically into an integrated chip containing many linear and nonlinear optical computing units, and used it to realize both an optical hidden Markov model and an optical recurrent neural network. The specific innovation is therefore an asynchronous on-chip recurrent accelerator in which wavelength relay removes one of the main timing bottlenecks of fast recurrent photonic hardware.

1.3.11 Reconfigurable versatile integrated photonic computing chip

Authors: Yufei Wang et al.
Journal: eLight 5, 20 (2025)
DOI: 10.1186/s43593-025-00098-6

Innovation, working principle, and use. This work presented a reconfigurable integrated photonic computing chip designed to support multiple neural architectures rather than a single fixed workload. By combining algorithm-hardware co-design with flexible microring-based building blocks and a rich optical source configuration, the chip can be reprogrammed across fully connected, convolutional, and recurrent computing patterns. The paper is important because it treats programmability as a first-class goal for photonic AI hardware. That makes it closer to a general photonic computing substrate than to a one-off accelerator for a single task.

1.3.12 Hardware-efficient photonic tensor core: accelerating deep neural networks with structured compression

Authors: Shupeng Ning et al.
Journal: Optica 12(7), 1079–1089 (2025)
DOI: 10.1364/OPTICA.559604

Innovation, working principle, and use. The tensor core is co-designed with structured model compression, so the photonic hardware spends channels only on the parts of the weight tensor that remain after pruning or factorization. This paper developed a hardware-efficient photonic tensor core that uses structured compression to reduce the size and control cost of optical neural computation. Instead of mapping dense weights directly, the architecture exploits compressed structure so the same photonic hardware can realize a larger effective model with fewer active elements. The idea is important because photonic accelerators often hit a wall in component count, control wires, and I/O before they hit a theoretical throughput limit. Structured compression therefore offers an appealing algorithm-hardware path to practical scaling.

1.3.13 Integrated optical-to-optical gain in a silicon photonic modulator neuron

Authors: Joshua C. Lederman et al.
Journal: Optica 12(12), 1904–1911 (2025)
DOI: 10.1364/OPTICA.567002

Innovation, working principle, and use. One major reason deep optical neural networks are hard to cascade is that every layer tends to lose signal power. This paper therefore focused on optical-to-optical gain at the level of a single photonic neuron. The neuron still uses the familiar modulator-neuron idea—weighted optical inputs are summed, converted to an electrical drive, and used to control a modulator—but the device is engineered so the resulting optical output can exhibit net gain relative to the optical input. That property is crucial because a useful multilayer network needs one stage to drive the next without continuously inserting bulky regeneration hardware. The specific innovation of the paper is to show that a silicon photonic modulator neuron can be made genuinely cascadable by integrating gain into the optical-to-optical transfer itself.

1.3.14 Integrated photonic 3D tensor processing engine

Authors: Yue Wu et al.
Journal: Light: Science & Applications 15, 154 (2026)
DOI: 10.1038/s41377-026-02183-y

Innovation. The paper solves a dataflow problem that ordinary photonic matrix-vector accelerators do not solve. Standard 2D photonic engines can multiply matrices quickly, but 3D convolution still needs large electrical overhead for tensor reshaping, channel synchronization, temporary storage, and digital accumulation. The 3D-TPE reorganizes the whole pipeline so those steps happen optically.

Working principle. The engine has three main parts arranged as OMU–OCU–OMU. The first optical memory unit (OMU) is a tunable delay-line chip. It receives the sequentially modulated input data, duplicates them across wavelength and spatial paths, and applies delays of $0, \Delta t, 2\Delta t, \ldots$ so the symbols that belong to one 3D kernel window arrive at the computing chip at the right relative times. The optical computing unit (OCU) is a dual-coupled-MRR crossbar. Its weighting elements store the 3D kernel values, and each weighting element handles one wavelength channel. Because the dual-coupled ring has a flat-top response and a 50 GHz 3-dB bandwidth, it can weight high-speed signals with less distortion than a single-ring filter. After the first weighted products are generated, the second OMU adds another set of delays so all products that belong to one 3D convolution output land in the same sampling window. A photodetector and electrical combiner then add the terms, so the sampled waveform is the convolution result. The main point is that wavelength, time, and space are interleaved together, so caching, synchronization, weighting, and accumulation stay inside the optical pipeline.

How to use it. To use the 3D-TPE, map the input tensor slices onto multi-wavelength optical carriers, choose OMU delays to match the symbol rate, load the 3D kernel values onto the crossbar weights, and stream the tensor once through the engine. The output waveform is then sampled at the correct interval to recover the 3D feature map. In the experiments, the system performed four-channel matrix multiplication from 10 to 30 Gbaud and executed the convolution layer of a 3D CNN for LiDAR point-cloud recognition at 20 Gbaud, reaching 97.06% classification accuracy.

1.3.15 Compact, reconfigurable, and scalable photonic neurons by modulation-and-weighting microring resonators

Authors: Weipeng Zhang et al.
Journal: eLight 6, 6 (2026)
DOI: 10.1186/s43593-026-00122-3

Innovation. The central idea is to let one microring do two jobs at once: modulate the input signal and provide the weight. That removes the usual need for a separate modulator ring and weighting ring, which makes the neuron smaller, easier to align spectrally, and much more energy efficient. The same ring bank can also be reconfigured into feedforward or recurrent neurons.

Working principle. Each ring contains a PN-doped waveguide for fast carrier-based modulation and an integrated metal heater for slow thermal tuning. The heater sets the operating point on the Lorentzian resonance curve, and the local slope at that operating point determines the effective small-signal weight. The PN junction then writes the input signal onto that slope. Because the same resonance curve is also nonlinear, changing the bias point changes not only the weight but also the activation shape, so the ring can produce sigmoid-like, ReLU-like, or other nonlinear responses. In the demonstrated 10-ring bank, adjacent rings are spaced by about 51 GHz so each wavelength addresses one ring. The drop-port signals are summed on one photodetector and transimpedance amplifier. For temporal tasks, the detected output can be routed back electrically to one or more rings with a chosen delay, which gives short-term or long-term memory without adding another photonic loop.

How to use it. In a feedforward mode, assign one wavelength per ring, set the heater currents to the desired weights, drive the PN junctions with the input streams, and detect the weighted sum at the output. In a recurrent mode, reserve one or more rings for delayed feedback and feed the detected output back into them through a controlled electrical delay path. The paper used nine rings as a 3×3 optical convolution kernel for image processing with errors below 5%, and it also used recurrent configurations for high-frequency financial time-series prediction. Each modulation-and-weighting element occupied only $80 \times 45\ \mu\text{m}^2$ and consumed an average of 0.186 mW.

1.4 In-memory computing

The common idea here is to store the weight inside the optical device itself, usually a phase-change cell, so memory and multiplication happen in the same physical location. That is the key difference from ordinary photonic processors, where a separate controller programs the weights. The entries below therefore focus on what is stored, how the stored state changes transmission, and how that stored transmission is used in a computation.

1.4.1 Parallel convolutional processing using an integrated photonic tensor core

Authors: Johannes Feldmann et al.
Journal: Nature 589(7840), 52–58 (2021)
DOI: 10.1038/s41586-020-03070-1

Innovation, working principle, and use. Instead of implementing only dense matrix multiplication, the tensor core is scheduled so one physical optical engine directly carries out several convolution kernels in parallel. This work introduced an integrated photonic tensor core for convolutional neural processing using phase-change-material weight elements on a silicon photonic platform. The architecture exploited parallel optical pathways and nonvolatile weight programming to execute convolution kernels directly in hardware, reducing the need for repeated weight loading. By combining integrated interferometric routing with in-memory photonic weighting, the paper showed that convolutional operations can be implemented as a compact programmable photonic primitive suitable for inference-oriented AI hardware.

1.4.2 Electrically programmable phase-change photonic memory for optical neural networks with nanoseconds in situ training capability

Authors: Maoliang Wei et al.
Journal: Advanced Photonics 5(4), 046004 (2023)
DOI: 10.1117/1.AP.5.4.046004

Innovation, working principle, and use. This paper is important because it brings nonvolatile optical weights closer to true trainable hardware. The device integrates a phase-change memory cell with a photonic circuit so the optical transmission or resonance encodes the synaptic weight, while an electrical pulse writes the new state. The key advance over slower nonvolatile photonic memories is nanosecond-scale electrical programming, which is fast enough to make repeated weight updates during hardware training thinkable rather than purely static. In other words, the cell is not only a memory that stores a finished model; it is a memory that can be updated quickly while remaining nonvolatile between updates. That combination—electrical programmability, optical readout, multi-level storage, and nanosecond writing—is the specific innovation that makes the paper especially relevant for optical neural networks.

1.4.3 Sparse weight mapping and computation reuse strategy for scalable photonic matrix multiplication

Authors: Cen Liao et al.
Journal: Optics Express 34(3), 4228–4241 (2026)
DOI: 10.1364/OE.584125

Innovation. The main contribution is a mapping method, not a new physical crossbar cell. The authors recognize that many useful convolution kernels are sparse and repetitive, so a small crossbar should not waste one physical weight cell on every matrix entry. Their sparse-weight-mapping and

computation-reuse strategy compresses the kernel before it is loaded onto hardware and then reuses partial results, allowing a 4×4 photonic crossbar to execute larger 3×3 filters.

Working principle. The hardware is a PCM photonic crossbar. Horizontal directional couplers broadcast the input optical power along each row, a GST cell at each crossing attenuates the local light according to its crystallization state, and vertical couplers collect the weighted signals into output columns. A column therefore performs a multiply–accumulate of the row inputs. The SWM-CR strategy reduces the hardware requirement in two steps. First, it removes zero-valued kernel entries, so computation scales with the number of nonzero coefficients instead of the full kernel size. Second, it merges repeated nonzero values, so the hardware only needs to store the unique coefficient values. The repeated use of the same coefficient is handled by regrouping the input streams in time and by reusing already computed partial results. For example, a vertical Sobel kernel can be rewritten using only the coefficient set $[-1, 1, 2, -2]$. Those four values are programmed into one column of the 4×4 crossbar, while the input channels are reorganized as sums such as $(I_1 + I_3)$ and $(I_7 + I_9)$ so the column output reconstructs the full 3×3 convolution.

How to use it. To use this design, first inspect the target kernel, keep only the distinct nonzero weights, and program those values into the PCM cells. Then regroup the input pixels that share the same coefficient, send those grouped signals into the crossbar using time-division multiplexing, and finally recombine or reuse the optical outputs in simple electrical post-processing to recover the full filter result. With this method, the fabricated 4×4 array achieved 3-bit modulation, executed four 3×3 edge-detection kernels with a 225% improvement in array utilization, and supported an MNIST CNN simulation with 96.7% accuracy.

1.4.4 Integrated all-photonic non-volatile multi-level memory

Authors: C. Rios et al.
Journal: Nature Photonics (2015)
DOI: 10.1038/nphoton.2015.182

Innovation, working principle, and use. Programming and reading are both optical, so the device behaves as a true photonic memory element rather than as an electronic memory that merely controls optics. This influential paper demonstrated a fully photonic nonvolatile multilevel memory in which optical pulses both programmed and read the stored state. The key advance over earlier binary cells was controlled intermediate phase states, which is essential for analog weighting and later in-memory computing.

1.4.5 Mixed-mode operation of hybrid phase-change nanophotonic circuits

Authors: Y. Lu et al.
Journal: Nano Letters (2017)
DOI: 10.1021/acs.nanolett.6b03688

Innovation, working principle, and use. The same PCM device can be written electrically or optically and read out optically, which makes it a bridge between electronic control and optical computation. The authors explored hybrid phase-change nanophotonic circuits that could be driven in mixed optical and electrical modes. That flexibility matters for in-memory computing because it separates high-bandwidth optical readout from more practical electrical programming, a combination that became common in later photonic memory chips.

1.4.6 Calculating with light using a chip-scale all-optical abacus

Authors: J. Feldmann et al.
Journal: Nature Communications (2017)
DOI: 10.1038/s41467-017-01506-3

Innovation, working principle, and use. The PCM cells are not only storing data; repeated partial switching lets the device accumulate analog optical contributions, which is why the authors compare it to an abacus. This paper showed that phase-change photonic devices can do arithmetic-like accumulation directly in the optical domain, effectively functioning as an all-optical abacus. It is an important conceptual bridge from photonic memory to photonic in-memory computing because stored material states directly participate in computation.

1.4.7 Device-Level Photonic Memories and Logic Applications Using Phase-Change Materials

Authors: Z. Cheng et al.
Journal: Advanced Materials (2018)
DOI: 10.1002/adma.201802435

Innovation, working principle, and use. This paper moves one level down to the device physics, explaining how the state-dependent optical constants of PCM cells can be harnessed for both memory and logic primitives. This paper developed pulse-width-modulated optical programming of phase-change photonic devices and showed device-level memory and logic functionality on an integrated platform. The main contribution was practical multilevel programming methodology, not just a memory cell, which made later photonic in-memory operations more feasible.

1.4.8 Fast and reliable storage using a 5 bit, nonvolatile photonic memory cell

Authors: X. Li et al.
Journal: Optica (2019)
DOI: 10.1364/OPTICA.6.000001

Innovation, working principle, and use. The emphasis is multilevel programming fidelity: one cell is driven to several stable intermediate states so it can store more than a binary optical weight. This paper demonstrated a 5-bit photonic memory cell, highlighting that multilevel non-volatile storage in integrated photonics can reach a resolution useful for analog computing. The work is important because it directly targets storage precision, which is one of the main barriers to photonic in-memory dot products and neural inference.

1.4.9 Miniature multilevel optical memristive switch using phase change material

Authors: H. Zhang et al.
Journal: ACS Photonics (2019)
DOI: 10.1021/acsphotonics.9b00819

Innovation, working principle, and use. The optical transmission of the device depends on its programming history, so it behaves like an optical memristor rather than like a memory cell with only two fixed states. The authors proposed and demonstrated a miniature optical memristive switch in which the optical response depends on the programmed history of the phase-change material. This memristive interpretation is attractive for in-memory computing because it links memory and weighting in one physical element.

1.4.10 Plasmonic nanogap enhanced phase-change devices with dual electrical-optical functionality

Authors: N. Farmakidis et al.
Journal: Science Advances (2019)
DOI: 10.1126/sciadv.aaw2687

Innovation, working principle, and use. The plasmonic nanogap concentrates light and current into a very small volume, dramatically lowering the switching energy of the phase-change region while allowing both electrical and optical access. This paper combined plasmonic nanogaps with phase-change media to realize devices that can be programmed electrically and interrogated optically. That dual electrical-optical functionality is highly relevant to photonic in-memory computing because practical systems often need electronic control with photonic readout.

1.4.11 In-memory computing on a photonic platform

Authors: C. Rios et al.
Journal: Science Advances (2019)
DOI: 10.1126/sciadv.aau5759

Innovation, working principle, and use. This is one of the foundational papers for photonic in-memory computing. The central idea is simple but powerful: instead of storing a weight in one device and then moving it to a separate optical processor, the storage element itself becomes the weight inside the optical computation. The authors used nonvolatile phase-change cells integrated with photonic waveguides so that the programmed transmission of each cell represented a synaptic coefficient. When an optical input passed through an array of such cells, each channel was multiplied by its stored transmission, and the weighted signals could then be summed to form a dot product. For a beginner, this is the optical equivalent of avoiding the memory wall in electronics: memory and computation are co-located in the same physical device. That is why the paper became a landmark in the phase-change route to optical computing.

1.4.12 All-optical spiking neurosynaptic networks with self-learning capabilities

Authors: J. Feldmann et al.
Journal: Nature (2019)
DOI: 10.1038/s41586-019-1157-8

Innovation, working principle, and use. The PCM weights are embedded in a spiking optical network where optical pulses act like spikes and repeated programming events strengthen or weaken synaptic transmission in a learning-like way. This paper used photonic phase-change devices as synapses in an all-optical spiking neurosynaptic network and showed learning-relevant behavior directly in hardware. Although it is often grouped with neuromorphic photonics, it is also a strong photonic in-memory computing example because the synaptic state is stored and used in the same optical device.

1.4.13 Photonic in-memory computing primitive for spiking neural networks using phase-change materials

Authors: I. Chakraborty et al.
Journal: Physical Review Applied (2019)
DOI: 10.1103/PhysRevApplied.11.014063

Innovation, working principle, and use. The paper isolates a single photonic building block for spiking networks, showing how one PCM-weighted optical path can implement the weighted accumulation required by spike processing. This paper formulated a photonic in-memory computing primitive specifically for spiking neural networks using phase-change materials. It is important because it mapped photonic memory cells to a concrete computational model rather than treating them only as generic tunable devices.

1.4.14 Tunable volatility of Ge2Sb2Te5 in integrated photonics

Authors: N. Youngblood et al.
Journal: Advanced Functional Materials (2019)
DOI: 10.1002/adfm.201807571

Innovation, working principle, and use. By controlling how quickly the material relaxes after programming, the same PCM family can behave either as a long-term memory or as a short-term volatile synapse. This paper studied how the effective volatility of GST can be tuned in integrated photonic devices, which is useful for balancing stable memory against transient response. That design freedom is interesting for compute hardware because some operations benefit from true nonvolatility while others need short-term plasticity.

1.4.15 Broadband transparent optical phase change materials for high-performance nonvolatile photonics

Authors: Y. Zhang et al.
Journal: Nature Communications (2019)
DOI: 10.1038/s41467-019-12196-4

Innovation, working principle, and use. The materials contribution is important because common PCM such as GST adds too much loss; a more transparent PCM lets the chip keep nonvolatile programmability without destroying optical throughput. This influential materials paper introduced broadband transparent optical phase-change materials that dramatically reduce loss compared with classical GST-like media. It is highly relevant to photonic in-memory computing because lower-loss nonvolatile materials remove one of the biggest obstacles to scaling cascaded memory-weight cells.

1.4.16 Experimental investigation of silicon and silicon nitride platforms for phase-change photonic in-memory computing

Authors: X. Li et al.
Journal: Optica (2020)
DOI: 10.1364/OPTICA.379228

Innovation, working principle, and use. The paper compares two host photonic platforms to see how the same PCM concept behaves when integrated on waveguides with different confinement, loss, and thermal properties. This study directly compared silicon and silicon nitride as host platforms for phase-change photonic in-memory computing. The paper is valuable because it moved the discussion from isolated device success to platform selection, showing how material stack and waveguide choice affect loss, programming, and usable computational precision.

1.4.17 Nonvolatile Electrically Reconfigurable Integrated Photonic Switch Enabled by a Silicon PIN Diode Heater

Authors: J. Zheng et al.
Journal: Advanced Materials (2020)
DOI: 10.1002/adma.202001218

Innovation, working principle, and use. A local PIN-diode heater provides a compact electrical method for writing a nonvolatile optical switching state into the PCM-assisted photonic device. This paper showed an electrically reconfigurable nonvolatile photonic switch using an integrated silicon PIN diode heater, a very practical programming approach for chip-scale systems. It demonstrated that photonic memory can be made more CMOS-compatible while still exploiting phase-change nonvolatility.

1.4.18 Chalcogenide phase-change devices for neuromorphic photonic computing

Authors: J. Brueckerhoff-Plueckelmann et al.
Journal: Journal of Applied Physics (2021)
DOI: 10.1063/5.0042549

Innovation, working principle, and use. This work frames chalcogenide PCM elements explicitly as neuromorphic photonic synapses and neurons, linking material switching dynamics to learning functions. This paper discussed and demonstrated how chalcogenide phase-change devices can support neuromorphic photonic computing. It is relevant here because it frames photonic memory devices as computational primitives rather than only reconfigurable optics, connecting stored states directly to weighted optical processing.

1.4.19 Programmable phase-change metasurfaces on waveguides for multimode photonic convolutional neural network

Authors: C. Wu et al.
Journal: Nature Communications (2021)
DOI: 10.1038/s41467-020-20365-z

Innovation, working principle, and use. The programmable metasurface perturbs several guided modes at once, so the multimode waveguide itself becomes a convolution engine whose weights are stored nonvolatilely in the phase-change pattern. This paper used programmable phase-change metasurfaces on waveguides to implement convolution-related functionality for photonic neural processing. It is an important in-memory-style result because the programmed phase pattern stays stored in the optical hardware and directly performs the operation during inference.

1.4.20 In-memory photonic dot-product engine with electrically programmable nonvolatile memories

Authors: H. Zhou et al.
Journal: Nature Communications (2023)
DOI: 10.1038/s41467-023-38473-x

Innovation, working principle, and use. This paper is a direct demonstration of the photonic dot-product engine that earlier memory papers were pointing toward. Each synaptic weight is stored in an electrically programmable nonvolatile optical memory cell, so the matrix coefficients remain on chip without static tuning power. Input data are injected optically, each channel is multiplied by the stored transmission of its memory cell, and the weighted optical signals are summed

by the photonic readout to produce the dot-product output. The important distinction is that the memories are not passive configuration elements in the background; they are the active weight bank of the arithmetic engine itself. This makes the work a very clear example of true photonic in-memory computing: programmable nonvolatile storage and matrix multiplication occur in the same photonic fabric.

1.4.21 Neuromorphic Photonic Memory Devices Using Ultrafast, Non-Volatile Phase-Change Materials

Authors: C. Chen et al.
Journal: Advanced Materials (2023)
DOI: 10.1002/adma.202203909

Innovation, working principle, and use. The focus is on materials and device design that preserve nonvolatility while switching fast enough for neuromorphic update rules and high-throughput optical operation. This paper focused on ultrafast nonvolatile phase-change materials for neuromorphic photonic memory devices. It is relevant because it targets the speed-memory trade-off directly and shows a path toward photonic memory elements that update quickly enough for practical adaptive hardware.

1.4.22 High-speed and energy-efficient non-volatile silicon photonic memory based on heterogeneously integrated memresonator

Authors: E. Tossoun et al.
Journal: Nature Communications (2024)
DOI: 10.1038/s41467-024-44773-7

Innovation, working principle, and use. This paper reported a high-speed and energy-efficient nonvolatile silicon photonic memory based on a heterogeneously integrated memresonator. The main contribution is the push toward more practical memory metrics while keeping compatibility with silicon photonics, which is crucial for scalable in-memory hardware.

1.4.23 All-optical organic photochemical integrated nanophotonic memory: low-loss, continuously tunable, non-volatile

Authors: R. Bilodeau et al.
Journal: Optica (2024)
DOI: 10.1364/OPTICA.529336

Innovation, working principle, and use. Instead of relying on a heavily absorbing chalcogenide PCM, the device uses a photochemical material whose refractive state can be tuned continuously with lower insertion loss. This paper introduced an organic photochemical nanophotonic memory with low loss, continuous tunability, and nonvolatility. It is exciting for in-memory computing because it expands the design space beyond phase-change crystallization and explicitly targets the analog low-loss regime that photonic weights require.

1.4.24 Reconfigurable inverse-designed phase-change photonics

Authors: C. Wu et al.
Journal: APL Photonics (2025)
DOI: 10.1063/5.0234637

Innovation, working principle, and use. This paper combined inverse design with pixel-level phase-change programming, allowing one photonic device to be reconfigured into different functions through stored nonvolatile patterns. That concept is highly relevant to in-memory computing because the memory state becomes part of the device geometry-like functionality itself.

1.4.25 High-speed multifunctional photonic memory on a foundry-processed photonic platform

Authors: S. Rahimi Kari et al.
Journal: Optica (2025)
DOI: 10.1364/OPTICA.536866

Innovation, working principle, and use. This paper is notable because it reported high-speed multifunctional photonic memory on a foundry-processed platform, making the field feel more manufacturing-oriented. The combination of speed, multiple functions, and foundry compatibility is exactly what photonic in-memory computing needs to move beyond laboratory prototypes.

1.5 Optical nonlinear activation function

These papers focus on the nonlinear step after a linear optical layer. The important question is not only whether a nonlinearity exists, but how the response curve is created, tuned, and inserted between layers without destroying the speed or energy advantage of optics. Each entry therefore emphasizes the physical source of the nonlinearity and how it is used in a neural-network pipeline.

1.5.1 Programmable low-threshold optical nonlinear activation functions for photonic neural networks

Authors: Ying Huang et al.
Journal: Optics Letters 47(7), 1810–1813 (2022)
DOI: 10.1364/OL.451287

Innovation, working principle, and use. The device aims to trigger a strong nonlinear response at a lower optical power, which is critical if many activation blocks must be cascaded in one network. This paper experimentally demonstrated programmable optical nonlinear activation functions with a low activation threshold, an important requirement for cascaded photonic neural computation. The device strategy used integrated photodetection and optical control to realize reconfigurable activation responses while avoiding the very high power levels that often hinder all-optical nonlinearities. Its main value lies in showing that practical, low-threshold, PIC-compatible nonlinear activation can be engineered rather than treated as a fixed material limitation.

1.5.2 Reconfigurable nonlinear photonic activation function for photonic neural network based on non-volatile opto-resistive RAM switch

Authors: Zefeng Xu et al.
Journal: Light: Science & Applications 11, 288 (2022)
DOI: 10.1038/s41377-022-00976-5

Innovation, working principle, and use. A nonvolatile opto-resistive memory switch is used to set the activation curve itself, so the nonlinear function can be reprogrammed and retained without continuous power. This paper introduced a reconfigurable photonic activation function based on a nonvolatile opto-resistive RAM switch. The central idea was to use a memory-like integrated device to program nonlinear optical responses without relying on continuous power-hungry

bias control. That approach is attractive for photonic neural networks because both weights and activations benefit from nonvolatile programmability, and the paper demonstrated that activation tuning can also be merged with emerging photonic memory concepts.

1.5.3 All-optical ultrafast ReLU function for energy-efficient nanophotonic deep learning

Authors: Gordon H. Y. Li et al.
Journal: Nanophotonics 12(5), 847–855 (2023)
DOI: 10.1515/nanoph-2022-0137

Innovation, working principle, and use. The goal is to approximate a ReLU-like transfer curve entirely optically and at ultrafast speed, so the nonlinearity can keep pace with nanophotonic linear layers. This paper experimentally demonstrated an ultrafast all-optical ReLU function using a periodically poled thin-film lithium niobate nanophotonic waveguide. Because ReLU is one of the most common nonlinearities in deep learning, implementing it directly in optics is especially meaningful for moving beyond hybrid architectures. The main contribution is showing that a compact integrated nonlinear optical device can produce a practically relevant activation shape with very low energy and ultrafast response, rather than only a generic saturating nonlinearity.

1.6 Other methods

1.6.1 Perfect linear optics using silicon photonics

Authors: Miltiadis Moralis-Pegios et al.
Journal: Nature Communications 15, 5468 (2024)
DOI: 10.1038/s41467-024-49768-y

Innovation, working principle, and use. This paper deals with a foundational issue that affects almost every programmable photonic processor: the mathematics says a mesh should realize an exact linear transform, but the fabricated chip contains loss, phase offsets, and coupler errors. The authors showed how to recover effectively perfect linear optics from imperfect silicon-photonic components by combining decomposition theory with calibration and compensation procedures. The underlying idea is that one should not demand every individual component be ideal; instead, the processor should be programmed so the overall transfer matrix is ideal even though the local hardware is not. That perspective is extremely important for optical computing, because large matrix engines are only useful if arbitrary target transformations can still be realized after fabrication. So the specific innovation of the paper is a practical route from imperfect photonic building blocks to an almost ideal system-level linear operator.

1.6.2 Waveguide-multiplexed photonic matrix-vector multiplication processor using multiport photodetectors

Authors: Rui Tang et al.
Journal: Optica 12(6), 812–820 (2025)
DOI: 10.1364/OPTICA.552023

Innovation, working principle, and use. This paper shifts some of the intelligence of the matrix engine from the interferometer network to the detector itself. In a standard optical matrix-vector multiplier, many weighted optical channels have to be routed and combined before being read out, and that routing becomes cumbersome as the port count grows. Here the authors instead use waveguide multiplexing together with multiport photodetectors that can accept several

optical contributions in one receiver structure. The multiplication is still performed optically by modulating and weighting the input channels, but the accumulation is partly embedded in the detector architecture, which reduces the need for an excessively large coherent mesh. The specific innovation is therefore a detector-aware matrix-vector multiplier in which receiver design becomes part of the scaling strategy for photonic linear algebra.

2 Diffractive neural network

All papers in this section use the same core mechanism: a trained phase or amplitude pattern reshapes a propagating wavefront by diffraction. The weights are therefore built into passive optical layers rather than stored in electronic memory or phase shifters. To avoid repetition, I focus in each entry on what that paper adds on top of the basic D2NN idea: better readout, invariance, programmability, compactness, robustness, or a new application.

2.1 All-optical machine learning using diffractive deep neural networks

Authors: Xing Lin et al.
Journal: Science (2018)

Innovation, working principle, and use. This is the paper that introduced the D2NN concept to a wide audience. The network is a stack of passive diffractive layers, and each point on each layer acts like a trainable neuron whose phase or amplitude transmission is optimized in software. When an input image illuminates the first layer, diffraction causes every point on the wavefront to spread and interfere with many others, so the whole multilayer system performs a learned linear transformation through wave propagation alone. After training, the layers are fabricated and placed in sequence, and the final optical field is directed toward class-specific detector regions. The specific innovation is that the neural-network weights are not stored as electronic numbers or heater settings; they are literally built into the transmission pattern of passive diffractive surfaces, so inference happens at the speed of light with almost no power beyond illumination.

2.2 Fourier-space diffractive deep neural network

Authors: Tao Yan et al.
Journal: Physical Review Letters (2019)

Innovation, working principle, and use. The extra Fourier stage means the learned modulation operates in the spatial-frequency domain, so the network can manipulate low- and high-frequency content more directly than a plain stack of phase screens. This work reformulated diffractive optical inference in Fourier space, using lens-assisted propagation to make the optical operator more compact and better structured. The paper showed that inserting Fourier transforms between trained modulation planes can improve representational efficiency and simplify some all-optical image-processing tasks.

2.3 Class-specific differential detection in diffractive optical neural networks improves inference accuracy

Authors: Jinxin Li et al.
Journal: Advanced Photonics (2019)

Innovation, working principle, and use. The detector plane is split into positive and negative evidence regions for each class, and the decision is made from their difference rather than from a single bright spot. This paper improved readout design by replacing single positive detectors with class-specific differential detection. By comparing paired detector signals for each class, the system increases contrast and makes the optical decision process less fragile than intensity-only readout.

2.4 Analysis of diffractive optical neural networks and their integration with electronic neural networks

Authors: Deniz Mengu et al.
Journal: IEEE Journal of Selected Topics in Quantum Electronics (2019)

Innovation, working principle, and use. Instead of only showing a device demo, it asks what a passive diffractive stack can and cannot express mathematically, and where an electronic stage must be inserted to recover missing flexibility. This paper clarified the mathematical capabilities and limitations of multi-layer diffractive optical networks and discussed how they can be combined with electronics. It was important because it separated the appealing physical intuition of D2NNs from their actual linear-algebraic behavior and positioned hybrid optical-electronic pipelines as a practical route forward.

2.5 Performing optical logic operations by a diffractive neural network

Authors: Cheng Qian et al.
Journal: Light: Science & Applications (2020)

Innovation, working principle, and use. Here the input bits are encoded optically, the trained diffractive layers reshape the field, and distinct output ports correspond to Boolean answers such as AND, OR, or XOR. This paper showed that a trained diffractive network can implement optical logic functions rather than only image classification. A single metasurface-based optical system was trained to realize different Boolean operations, broadening the view of D2NNs from pattern recognition to general optical information processing.

2.6 Scale-, shift-, and rotation-invariant diffractive optical networks

Authors: Deniz Mengu et al.
Journal: ACS Photonics (2021)

Innovation, working principle, and use. The specific trick is to train the diffractive layers against transformed versions of the same object so the output detector pattern stays stable when the input moves, rotates, or changes size. This work built invariance to geometric transformations directly into diffractive network training. By exposing the model to random translations, rotations, and scale changes during optimization, the resulting optical network became far more usable for realistic machine-vision scenarios.

2.7 Ensemble learning of diffractive optical networks

Authors: Md Sadman Sakib Rahman et al.
Journal: Light: Science & Applications (2021)

Innovation, working principle, and use. Instead of trusting one passive classifier, the paper combines several independently trained optical predictors so their output votes average out individual failure modes. This paper imported the ensemble-learning idea into diffractive optics, combining multiple diffractive subnetworks or decisions to improve inference accuracy. The main contribution was to show that optical diversity can reduce the brittleness of single passive diffractive classifiers.

2.8 Computer-free, all-optical reconstruction of holograms using diffractive networks

Authors: Md Sadman Sakib Rahman and Aydogan Ozcan
Journal: ACS Photonics (2021)

Innovation, working principle, and use. The target is not only classification but inverse imaging: the trained layers convert a raw hologram into a cleaned reconstruction and suppress twin-image artifacts before any digital computation. This paper used a diffractive network to reconstruct in-line holograms and suppress twin-image artifacts without digital post-processing. It is important because it showed that diffractive networks can function as all-optical inverse solvers for imaging, not only as classifiers.

2.9 Space-efficient optical computing with an integrated chip diffractive neural network

Authors: Hao-Hui Zhu et al.
Journal: Nature Communications (2022)

Innovation, working principle, and use. This paper is important because it moved diffractive computing from a bulky free-space optical bench onto an integrated chip. The same basic principle remains: a trained phase distribution shapes the optical field, and propagation mixes the channels through diffraction. But instead of requiring large distances between separate free-space layers, the authors used a compact on-chip diffractive region so the spatial mixing happened inside an integrated photonic structure. That dramatically improves footprint, alignment stability, and the possibility of co-integration with other chip-scale optical components. The specific innovation is therefore the translation of D2NN-style diffraction from a laboratory free-space system into a space-efficient integrated-chip implementation.

2.10 A programmable diffractive deep neural network based on a digital-coding metasurface array

Authors: Chenliang Liu et al.
Journal: Nature Electronics (2022)

Innovation, working principle, and use. Because the metasurface pixels are digitally coded, the learned phase pattern is no longer frozen after fabrication; the same hardware can be repro-

grammed for different optical functions. This was a landmark reconfigurable diffractive paper: instead of a permanently fixed passive mask, the network used a digitally coded metasurface array whose states can be updated after fabrication. That makes one physical diffractive processor capable of switching functions or tasks, which is a major practical advance over one-shot printed layers.

2.11 Polarization multiplexed diffractive computing: All-optical implementation of a group of linear transformations through a polarization-encoded diffractive network

Authors: Jiawei Li et al.
Journal: Light: Science & Applications (2022)

Innovation, working principle, and use. Polarization serves as an extra label on the optical signal, allowing one physical diffractive stack to implement several different linear transforms depending on the input polarization state. This paper used polarization as an additional degree of freedom so one diffractive system can implement multiple linear transformations. It expanded diffractive computing capacity without simply stacking more physical layers or replicating the entire optical processor.

2.12 Advanced all-optical classification using orbital-angular-momentum-encoded diffractive networks

Authors: Kuo Zhang et al.
Journal: Advanced Photonics Nexus (2023)

Innovation, working principle, and use. Instead of ordinary pixel encoding, the network processes orbital-angular-momentum modes, so the diffractive layers learn to sort structured wavefronts carrying different azimuthal phase twists. This paper encoded information into orbital angular momentum channels and used diffractive networks to perform all-optical classification. The approach shows how D2NNs can benefit from structured-light encoding to enrich the information content entering the optical processor.

2.13 Quantitative phase imaging through random diffusers using a diffractive optical network

Authors: Yue Li et al.
Journal: Light: Advanced Manufacturing (2023)

Innovation, working principle, and use. The learned diffractive layers are trained as an optical inverse solver that undoes the scrambling caused by a random diffuser and recovers quantitative phase information at the output plane. This paper demonstrated that a learned diffractive optical network can help recover quantitative phase even when a random diffuser distorts the wavefront. It is an elegant example of using learned optics to undo hard inverse problems in a passive, front-end fashion.

2.14 Compact eternal diffractive neural network chip for extreme environments

Authors: Yilin Dong et al.
Journal: Communications Engineering (2024)

Innovation, working principle, and use. The emphasis here is physical robustness: the diffractive function is embedded in a compact chip intended to keep working where alignment-sensitive free-space systems would be impractical. This paper emphasized robust chip-scale diffractive computing for harsh or extreme environments. The main message is that nanoprinted or compact diffractive hardware can keep functioning where more delicate optical or electronic inference hardware may struggle.

2.15 Direct electromagnetic information processing with planar diffractive neural network

Authors: Zhaohui Gu et al.
Journal: Science Advances (2024)

Innovation, working principle, and use. The same idea is generalized beyond visible free-space optics to planar electromagnetic waves, showing that learned diffraction can be treated as an analog wave-computing primitive in broader frequency ranges. This paper extended the diffractive-neural idea to planar electromagnetic information processing, showing that the learned-metasurface concept can work well beyond the conventional optical image-classification framing. It is important because it links diffractive neural computation to broader wave-based signal processing.

2.16 Pluggable multitask diffractive neural networks based on cascaded metasurfaces

Authors: Cong He et al.
Journal: Opto-Electronic Advances (2024)

Innovation, working principle, and use. Different metasurface blocks can be inserted or removed like modules, so one optical front end can be reconfigured for several tasks without redesigning the entire stack. This work proposed a plug-and-play multitask diffractive system using cascaded metasurfaces. By fixing some layers and exchanging or reordering task-specific layers, it introduced a more modular view of diffractive optical computing.

2.17 All-optical image denoising using a diffractive visual processor

Authors: Çağatay Işıl et al.
Journal: Light: Science & Applications (2024)

Innovation, working principle, and use. The detector output is not a class label but a cleaned image, so the trained diffraction pattern acts like an optical denoising filter implemented by propagation rather than digital convolution. This paper used diffractive processing as an optical denoiser, learning front-end wave manipulation that suppresses noise before digital post-processing. It is a strong example of D2NNs serving as passive optical pre-processors for imaging rather than stand-alone classifiers.

2.18 Coupling Enhanced Diffractive Deep Neural Network with Structural Nonlinearity

Authors: Ouling Wu et al.
Journal: Advanced Photonics Research (2025)

Innovation, working principle, and use. This paper tries to go beyond a purely linear diffractive stack by strengthening optical coupling and introducing a structural nonlinearity so the physical network can represent richer mappings. This paper tried to boost D2NN expressivity by jointly exploiting stronger optical coupling and structural nonlinearity. The contribution is to show that useful nonlinear behavior can emerge from co-designing propagation and structure rather than relying only on an external nonlinear device.

2.19 Nanoprinted High-Neuron-Density Optical Linear Perceptrons Performing Near-Infrared Inference on a CMOS Chip

Authors: Elena Goi, Xi Chen, Qiming Zhang, Benjamin P. Cumming, Steffen Schoenhardt, Haitao Luan, and Min Gu
Journal: Light: Science & Applications 10, Article 40 (2021)
DOI: 10.1038/s41377-021-00483-z

Innovation, working principle, and use. This paper proposed a very different route from MZI-mesh or microring-based photonic neural networks: instead of using tunable interferometers, it built *passive diffractive optical linear perceptrons* as nanoprinted holographic elements directly on a CMOS chip. The core device is a machine-learning decryptor (MLD), which can be viewed as a single-layer diffractive neural network operating in transmission mode. During computer-based training, the phase delay of each artificial neuron in the diffractive layer is optimized by backpropagation so that a given input image is mapped to a desired optical output. After training, the learned phase profile is physically fabricated as a nanoscale 3D structure by *galvo-dithered two-photon nanolithography* (GD-TPN), which gives axial nanostepping down to 10 nm and enables a neuron density of more than 5×10^8 neurons/cm^2. The important point is that the trained diffractive element itself then performs the inference task *passively and unpowered*: when an input optical key propagates through the printed structure, diffraction and interference implement the learned linear transformation at the speed of light. The paper demonstrated this concept through optical decryption tasks. In the symmetric-decryption mode, only one specific input key retrieves the correct output, while incorrect keys are rejected. In the asymmetric-decryption mode, an entire class of keys can be recognized and mapped to a corresponding output, which is closer to flexible authentication. The authors also explored very compact multilayer designs with adjoining diffractive layers and argued that in such compact cases the system behaves like a tailored linear multiplexor rather than a conventional deep diffractive network with large interlayer spacing. Experimentally, the printed decryptors operated at 785 nm, were directly integrated on CMOS chips, and achieved all-optical inference with a working distance as small as 62.8 μm, highlighting an unusually compact on-chip free-space optical computing geometry.

2.20 Compact Optical Convolution Processing Unit Based on Multimode Interference

Authors: Xiangyan Meng, Guojie Zhang, Nuannuan Shi, Guangyi Li, José Azaña, José Capmany, Jianping Yao, Yichen Shen, Wei Li, Ninghua Zhu, and Ming Li

Journal: Nature Communications 14, 3000 (2023)
DOI: 10.1038/s41467-023-38786-x

Innovation, working principle, and use. This paper proposed a *compact on-chip optical convolution processing unit* (OCPU) that uses *multimode interference* rather than a large MZI mesh to perform convolution operations. The main motivation was scalability: many previous photonic computing schemes require a number of optical components that grows quadratically with matrix size, which makes large convolution layers difficult to integrate. In contrast, this work designed an *incoherent* silicon-nitride convolution unit whose component count scales *linearly* with the computational size. The basic 4-input OCPU consists of only *two 4×4 MMI cells and four phase shifters*, yet it can simultaneously implement *three 2×2 correlated real-valued kernels* in parallel. The input data are encoded onto four incoherent optical channels, sent through the cascaded MMI-plus-phase-shifter structure, and then detected by photodetectors so that each output port realizes a weighted sum of the four inputs. In this way, each row of the effective transfer matrix acts as one convolution kernel. The phase shifters thermally reconfigure the kernel values, so the same hardware can implement different convolution operators without changing the chip structure. Because direct square-law detection naturally gives non-negative kernel weights, the paper further introduced a differential strategy to realize *negative-valued kernels*: one output is treated as a reference ground line and subtracted from the others. Using this OCPU as the optical front-end and combining it with an electronic fully connected layer, the authors experimentally demonstrated ten-class MNIST handwritten-digit classification with *92.17%* accuracy, compared with *94.51%* in theory. The paper also reported that for non-negative kernels the convolution time per image was about *91.08 ns*, corresponding to roughly *10.98 million images per second*. Overall, the main contribution is not just another optical convolution demo, but a hardware architecture that reduces component count very aggressively while still providing dynamic reconfigurability and parallel kernel extraction.

2.21 Hundred-Layer Photonic Deep Learning

Authors: Tiankuang Zhou, Yizhou Jiang, Zhihao Xu, Zhiwei Xue, and Lu Fang
Journal: Nature Communications 16, 10382 (2025)
DOI: 10.1038/s41467-025-65356-0

Innovation, working principle, and use. This paper addressed one of the most fundamental bottlenecks in photonic neural networks: *depth*. Previous optical neural networks usually implement one logical linear layer through *multiple physical propagation stages* on chip, such as cascaded MZI meshes or multi-stage diffractive propagation. Because photonic computing is analog, every propagation stage and every nonlinear readout introduces noise and numerical deviation, so the overall error accumulates rapidly with depth. The authors argued that this error accumulation is not only a device-precision problem, but also a *redundancy problem*: conventional architectures use many correlated propagation stages to realize one matrix operation, and these redundant layers amplify analog error when networks become very deep. To solve this, the paper proposed a *single-layer photonic computing* chip, abbreviated as *SLiM*. The central idea is to realize an arbitrary matrix-vector multiplication using only *one physical propagation layer* instead of many cascaded optical layers. In the SLiM architecture, different inputs are assigned to different wavelengths and multiplexed into the chip. The target matrix is implemented through a modulated on-chip wavefront, but a direct mapping would normally fail because neighboring wavelengths and propagation channels are strongly correlated. The paper therefore introduced *on-chip perturbations* to decorrelate the propagation channels, so that the matrix-realization equations become linearly independent and arbitrary matrices can be solved and loaded onto the chip. In other words, rather than fighting correlation only through more precise calibration, the design intentionally breaks the correlation

structure of the hardware itself, which is the key conceptual advance of the paper. The paper further recognized that even if the linear propagation error is reduced, ordinary nonlinear activations can still spread noise from one layer to the next. To suppress this, the authors introduced a *detection-based perturbed activation* scheme. The practical implementation is not a purely optical nonlinearity, but a detection-and-ADC-based function that pushes distorted analog outputs toward the nearest valid signal levels, so that the error is decorrelated from the useful signal and does not continue to grow uncontrollably in subsequent layers. In this way, both the linear stage and the nonlinear stage are redesigned for *error tolerance*, not just for mathematical completeness. Based on this architecture, the paper claimed that the error rate can remain *bounded* even when the network depth exceeds 100 or 200 layers, instead of diverging with depth as in conventional ONNs. Experimentally, the authors built deep photonic networks and evaluated them at up to *10 GHz*. For image classification, they constructed a *100-layer residual network* and tested it on the full *ImageNet-1000* dataset. The reported photonic accuracy was *85.2%*, very close to the digital reference of *85.9%*. They also compared single-layer-chip ONNs with conventional multi-layer-chip ONNs under noise and showed that the single-layer scheme retained strong performance while the conventional deep optical networks collapsed rapidly as frequency and depth increased. Beyond image classification, the paper pushed the idea much further and experimentally evaluated large transformer-style models built on the same SLiM principle: a *0.345-billion-parameter, 384-layer* model for text generation and a *0.192-billion-parameter, 640-layer* model for conditional image generation. The reported photonic loss values remained close to digital references, with *3.04 vs. 2.96* for language generation and *7.32 vs. 7.28* for image generation. The paper therefore made a very ambitious claim: that by removing redundant propagation and making activation error-tolerant, photonic hardware can move from shallow optical inference toward *very deep* model execution, even for large-model-style workloads. Another notable point is that the architecture naturally supports *chip-to-chip* composition, so the spatial depth can extend from millimeter-scale on-chip propagation to *hundred-meter-scale* three-dimensional chip clusters, which the authors framed as a kind of photonic chiplet strategy for large AI systems.